



Statistical Thinking

Megan Ayers
Stat 141 | Fall 2026
Monday, Week 1

Goals for Today

1. Getting started in Stat 141
 - i. Course structure and technologies
 - ii. Where to find resources
 - iii. Course expectations
2. Introduce statistical thinking
3. Introduce data frames

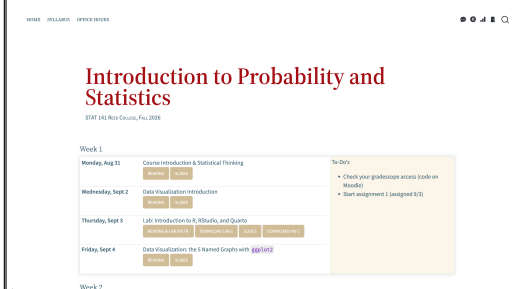
A bit about me

- Boise, ID → Lewis & Clark College (Portland) → Yale (New Haven) → back to Portland!
- My Research Interests:
 - Causal inference
 - Environmental policy evaluation
 - Text as data
 - R programming
- I'm looking forward to getting to know each other more in lab!

Getting Started in Stat 141

Getting Started in Stat 141

Course website



- The course website, megan-k-ayers.github.io/stat141-fall26, will be the central location for all our course materials.
- We'll use a few other resources to navigate Stat 141, but everything will be linked/directed to from the course website.

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Getting Started in Stat 141

Other Resources

Reed's [RStudio Server](#), for coursework,

A course-wide [Slack](#) workspace, for course communication,

[Gradescope](#), for turning in assignments, and

[Moodle](#), sparingly, for private course materials & info about meeting times/locations

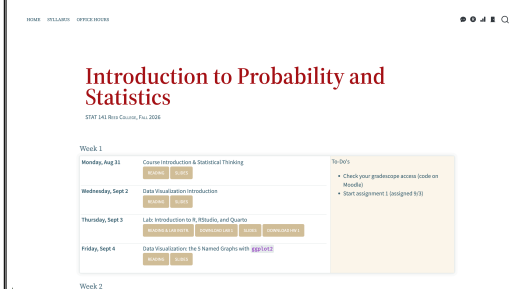
Textbooks (all free and online)

- [Statistical Inference via Data Science: A ModernDive into R and the tidyverse, Second Edition \(MD\)](#)
- [Introduction to Modern Statistics, Second Edition \(IMS\)](#)
- [OpenIntro Statistics, Fourth Edition \(OI\)](#)

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Getting Started in Stat 141

Finding Resources on the Course Website



- megan-k-ayers.github.io/stat141-fall26
- The syllabus and information about office hours are linked in the upper left
- Slack, RStudio, Gradescope, and Moodle are linked via icons in the upper right, in addition to a search button that covers all slides and readings!
- Readings, labs, homeworks, and slides will be posted next to the corresponding day

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Getting Started in Stat 141

Other Resources

- You will need access to a computer for this course.
 - **Please let me know ASAP after class or via email if you do not have access to a personal computer.**
- Attendance will be taken via QR code or link at the start of each lecture.
 - **Please let me know if you plan to be device-free during lectures so that I can accommodate this.**
- The internet in general is **not** a permitted resource, unless stated otherwise. If you feel additional resources are necessary for your learning, let's meet and discuss.

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My info

Name: Megan Ayers (she/her), you can call me 'Megan'

Email: meganayers@reed.edu

Office and office hours: Posted on Moodle, or by appointment

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Learning support resources

- The Stat 141 Teaching Team are excited to support your learning!
 - We'll have 2 course assistants holding additional office hours every week
- Each student is entitled to **one hour of individual tutoring** per week through the Office of Academic Support
 - This is an awesome and under-utilized resource!!

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A typical week in Stat 141

- **Monday**
 - Attend lecture
- **Wednesday**
 - Attend lecture
- **Thursday**
 - Submit homework assignment by 9am
 - Attend lab
 - Next homework assignment is released
- **Friday**
 - Attend lecture

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Assignments and Exams

- **Lab assignments (weekly)**
 - Assigned Thursday in lab
 - Due at the end of lab (+grace period for formatting issues) on **Gradescope**
 - Mostly collaborative R practice in small groups
- **Homework assignments (weekly)**
 - Assigned on Thursdays
 - Due the next Thursday at 9:00am on **Gradescope**
 - Builds on concepts from lecture and practice from lab. Please collaborate!
 - Always check your formatting!
- **Quizzes (weekly)**
 - Taken at the beginning of lab, in person and on paper
 - Short (30 mins) and low-stakes
 - Based closely on homework assignment
 - Lowest score dropped, revisions possible
- **NOTE:**
 - Office hours will be a great place to work through assignments and prepare for quizzes
 - Teaching team may indirectly or directly discuss solutions in OH or in feedback
 - You may help each other based on your own work, but **you may not distribute solutions** to students in this or other sections of 141

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Assignments and Exams

- **Exam (1 final exam)**
 - Cumulative, written, and in-person
- **Readings (each lecture, some labs)**
 - Opportunities to reinforce understanding of material
 - Similar content to lecture, varying presentation/examples

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Expectations for submitted work

- Homeworks are assessed for **completion** and **effort**.
 - Address all parts of questions completely whenever you can. If you have a partial solution, say so, explain what you've tried.
 - Respond in full sentences for written answers, and explain your reasoning.
 - Show your work, not just your final answer/value.
 - Code in submitted assignments should run (not be commented out whenever possible) and include a clear attempt to address the question.
- Quizzes are assessed for **accuracy**.
 - Similar expectations apply: respond in full sentences that address all parts of problems, explain your reasoning, show your work.

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Expectations for submitted work

- **Collaboration:**
 - An excellent way to deepen your understanding and enjoy the learning experience
 - You should try all problems on your own first
 - Can involve comparison of reasoning and solutions when each student has attempted a problem
 - Should **never** involve direct sharing or copying of solutions
 - Should be guided by the honor principle

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Late Work

Circumstances expected or unexpected may come up that result in missed or late submissions.

- **Labs:** each has a small but meaningful contribution towards your participation grade. No late submissions, but you may miss one lab without penalty.
- **HW:** up to 3 extensions days can be used throughout the semester.
 - e.g., 1 additional day for 3 HWs, 3 additional days for 1 HW, ...
 - Rounding days up
 - Further late work is accepted for half credit.
- **Quizzes:**
 - Lowest quiz score will be dropped (including if you missed a quiz).
 - Missed quizzes will receive scores of 0, and cannot be made up/revised
- **Exams:** no late exams are accepted. Please arrive on time!

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Engagement

- Being **actively present** is key. This requires **attendance**, and can look like:
 - Active listening
 - Contributing to discussion at the small group and/or classroom level
 - Contributing to discussions on Slack
- Again, I understand that things come up! You have 4 “free” lecture absences and 1 “free” lab absence. After this, a small, accumulating impact on participation grade.

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Engagement

Technology policy

*In STAT 141, we value a focused, distraction-free environment. Personal devices (including phones and headphones) and computers are generally not allowed in **lecture** to minimize digital distractions. However, technology is welcome for students who require digital tools for accessibility. We ask that these students limit their device use to essential in-class activities, such as note-taking or viewing digital readings. Non-academic use—including social media, messaging, and general web browsing—is not allowed, to ensure we remain fully engaged with the course material. Note that all lecture slides will be made available ahead of lecture on the course website.*

00:45

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Engagement

- I have high expectations but know that all of you (regardless of your stats, math, or computing background) have the ability to meet them.
- We are all going to make mistakes, we will learn more because of them.
- Resilience is one of the most valuable **skills** in learning.
- Expect some moments of repetition and try to lean in to the practice.

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Artificial Intelligence (AI) Policy

*Artificial intelligence (AI) tools, such as ChatGPT, Claude, Co-Pilot, Gemini, and others are being used to generate code, analyze data, and much more. However, learning to think critically about a problem at hand, building resilience in moments of struggle and confusion, and engaging with your peers, tutors, and instructors when not understanding a concept or question are integral components of a liberal arts education. Further, a key goal of this course is for you to learn how to thoughtfully, ethically, and **independently** extract knowledge from data and engage in statistical reasoning. Therefore, the use of generative AI tools, such as ChatGPT and others, are strictly prohibited in any stage of the work process for this course. The **Honor Principle** must guide your conduct in following this policy. If you have questions about whether a tool is allowed for this course, ask before using it.*

If you are unsure of how to approach a particular exercise on an assignment, or are stuck on a concept from lecture, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

01:15

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Course Climate

We expect everyone in this class to strive to foster a learning environment that is equitable, inclusive, and welcoming. If you experience any barriers to learning, please come to Professor Megan Ayers or a college administrator with your concerns.

Code of Conduct:

We expect all members of Stat 141 to make participation a harassment-free experience for everyone, regardless of age, body size, visible or invisible disability, ethnicity, sex characteristics, gender identity and expression, level of experience, education, socio-economic status, nationality, personal appearance, race, religion, or sexual identity and orientation.

We expect everyone to act and interact in ways that contribute to an open, welcoming, inclusive, and healthy community of learners. You can contribute to a positive learning environment by demonstrating empathy and kindness, being respectful of differing viewpoints and experiences, and giving and gracefully accepting constructive feedback.¹

1. This Code of Conduct is adapted from the [Contributor Covenant](#), version 2.0.

00:45

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Office Hours

- Office hours are an excellent opportunity to get help on homework assignments or concepts from lecture, and to collaborate with your peers
- The teaching team is more than happy to assist you, or just to chat and get to know you!

Important! Please fill out [the availability survey linked here](#) after lecture if you haven't already!!!

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Academic Accommodations

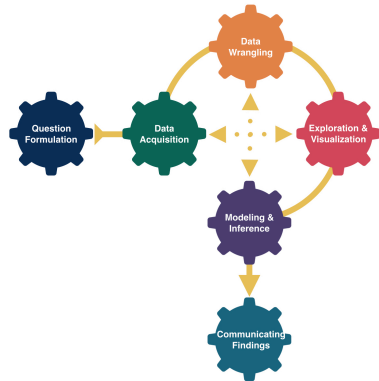
If you plan to request academic accommodations, please submit these **through the DAR student portal** as soon as possible.

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Course or syllabus questions?

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Stat 141: Learning Outcomes



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Stat 141: Learning Outcomes



- In this course, you will learn how to think critically with data by engaging in the entire data analysis process.
- Most of our time will be spent in the **Exploration and Visualization**, **Data Wrangling**, and **Modeling and Inference** steps, but we will spend some time in each cog!
 - First ~3 weeks in **Exploration and Visualization**, **Data Wrangling**, and **Data Acquisition**
 - Next ~2 weeks in **Modeling**
 - Next ~6 weeks of the course in **Inference**
 - Final weeks combining **Modeling** and **Inference**

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Statistical thinking

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Stat 141 is about developing our **statistical thinking** skills.

What is **statistical thinking**?

It is distinct from mathematical thinking.

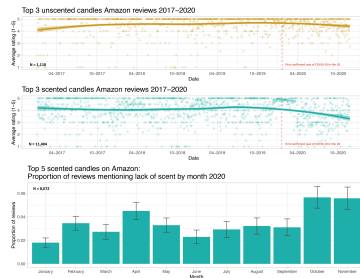
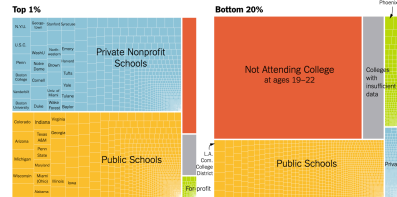
Let's discover what **statistical thinking** is through some examples.

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Data in Stat 141

- We'll motivate topics with a wide-range of **real** and **relevant** data examples.

Where the top 1% and the bottom 20% go to college



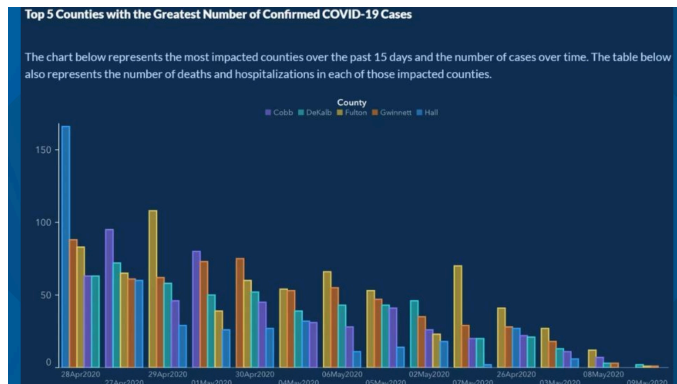
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Example: Visualizing COVID Prevalence

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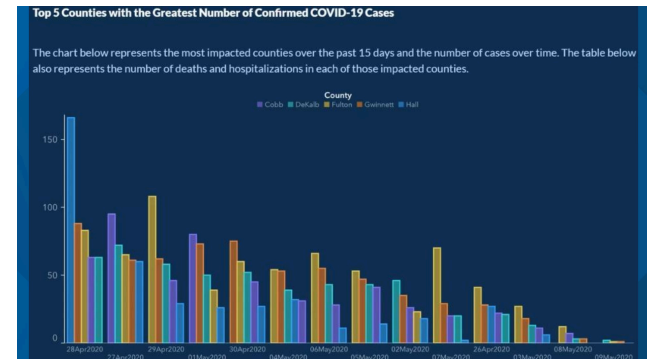
Example: Visualizing COVID Prevalence

- In May of 2020, the Georgia Department of Public Health posted the following graph:



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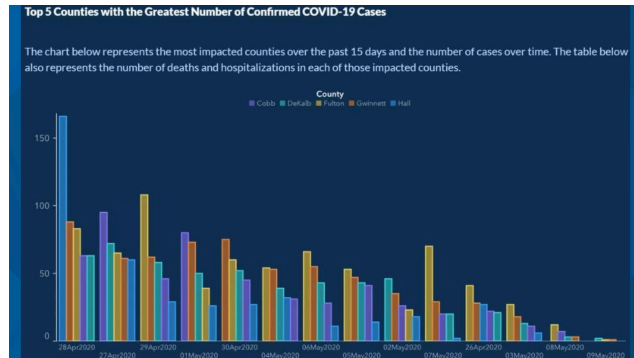
Example: Visualizing COVID Prevalence



- Q: At a quick first glance, what story does the graph appear to be telling?

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Example: Visualizing COVID Prevalence

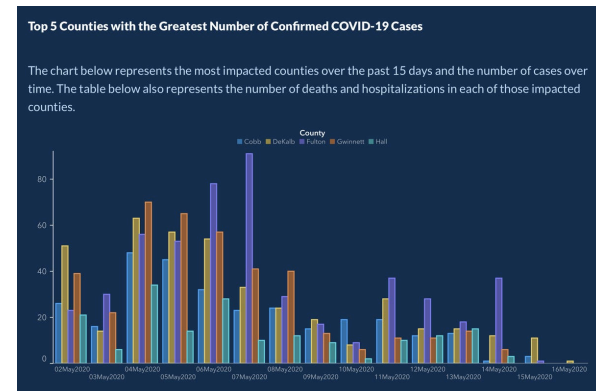


- Q: What is misleading about the graph? How could we fix this issue?

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Example: Visualizing COVID Prevalence

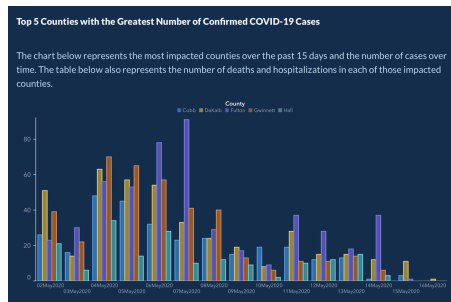
- After public outcry, the Georgia Department of Public Health updated the graph:



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Example: Visualizing COVID Prevalence

- After public outcry, the Georgia Department of Public Health updated the graph:

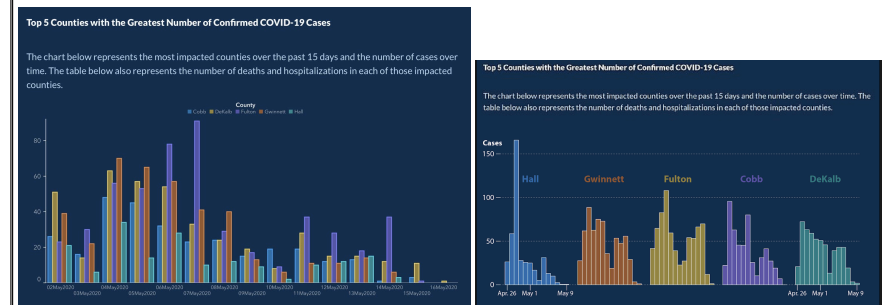


- Q: How do your conclusions about COVID-19 cases in Georgia change when interpreting the new graph?

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Example: Visualizing COVID Prevalence

Alberto Cairo, a journalist and designer, created the second graph of the Georgia COVID-19 data:



- A key principle of data visualization is to “help the viewer make meaningful comparisons”.
- Q: What comparisons are made easy by the lefthand graph? What about by the righthand graph?

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Statistical Thinking

- About developing **reasoning** (not just learning definitions and formulae).
- Statistical thinking requires **judgment** that takes time to develop.
 - Will see **examples** and **practice** applying statistical thinking throughout the course.
 - Numeric calculations only get us so far: it is critical to understand the **underlying data** and **assumptions**
- Developing our statistical thinking skills will allow us to soundly **extract knowledge from data**!

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What are/is Data?

“‘Raw data’ is an oxymoron.” – Lisa Gitelman

“Data ... is information made tractable.” – Catherine D’Ignazio and Lauren Klein

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Data Frames

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Data Frames

Data in **spreadsheet**-like format where:

- Rows = Observations/cases
- Columns = Variables

ID	kind	.pred_AI	.pred_class	detector	native	name	model
1	Human	0.9999942	AI	Sapling	No	Real TOEFL	Human
2	Human	0.8281448	AI	Crossplag	No	Real TOEFL	Human
3	Human	0.0002137	Human	Crossplag	Yes	Real College Essays	Human
4	AI	0.0000000	Human	ZeroGPT	NA	Fake CS224N - GPT3	GPT3
5	AI	0.0017841	Human	OriginalityAI	NA	Fake CS224N - GPT3, PE	GPT4
6	Human	0.0001783	Human	HFOpenAI	Yes	Real CS224N	Human

- Data from **GPT Detectors Are Biased Against Non-Native English Writers**. *Weixin Liang, Mert Yuksekgonul, Yining Mao, Eric Wu, James Zou*. **CellPress Patterns** and available in the **R** package **detectors**.

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Data Frames

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Rows = Observations/cases

What are the cases? What does each row represent?

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Data Frames

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Columns = Variables

Variables: Describe characteristics of the observations

- **Quantitative:** Numerical in nature
- **Categorical:** Values are categories
- **Identification:** Uniquely identify each case

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Data Frames

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Every time you get a new dataset, spend time exploring the variables.

Example questions:

- Is each variable capturing the desired information?
- For categorical variables, what are the categories? Do those categories adequately represent that variable?
- For quantitative variables, what values are possible? Were the data rounded or binned? Are those values actually encoding categories? What are the units of measurement?

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Recap from today

- Course resources, logistics, policies, expectations
- Introduced big picture of statistical thinking
- Introduced data frames

Next time

- Introduction to data visualization!

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